CHAPTER 25

Venous Malformations

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Vascular malformations are likely the single most misdiagnosed entity in the vascular system. Essentially, vascular malformations are errors in vasculogenesis with the particular characteristics of the lesion determined by the vessel in the vascular system that is involved. As a result, these malformations can include arteries, veins, lymphatic vessels, or capillaries. These lesions occur in about 1.5% of the population and over 90% are present at birth. Venous malformations are the most common vascular anomalies. Although these lesions can have mass effect on adjacent structures, they are not tumors.

An organized, histologically based classification system was initially presented by Mullikan, Glowacki, and colleagues in 1992, and a more recently modified classification system was adopted in 2014 by the International Society for the Study of Vascular Anomalies (ISSVA). The classification system clearly separates tumors (e.g., hemangiomas) from the vascular spaces that characterize a vascular malformation[1] (Table 25.1).

WHAT ARE HEMANGIOMAS? ARE THEY ARTERIOVENOUS MALFOMATIONS?

Hemangiomas are childhood masses characterized histologically by high endothelial cell turnover and by cell markers (GLUT-1, merosin, Lewis Y) that are traditionally found only in human placental tissue and that display phasicity characterized by a rapid proliferative phase, plateau phase, and slow involutional phase. The proper nomenclature for these lesions is *infantile hemangioma*. Infantile hemangiomas are benign vascular tumors that are not usually present at the time of birth but instead become evident

Abstract

Vascular malformations are errors in vasculogenesis with the particular characteristics of the lesion determined by the vessel in the vascular system that is involved. As a result, these malformations can include arteries, veins, lymphatic vessels, or capillaries. Hemangiomas are NOT arteriovenous malformations (AVMs), and vice versa. Low-flow venous malformations are the most common form of vascular malformation—simplified, they are a number of tortuous vascular channels. Arteriovenous fistulas (AVFs) are an abnormal connection between an artery and a vein. Although these high-flow lesions can be congenital, often they are acquired as the result of surgery, penetrating trauma, or erosion by adjacent disease processes. The Yakes' classification is a relatively new system that characterizes AVMs based on their angioarchitectures suggesting possible treatment strategies. Treatment of venous malformations has improved over the past decade as a result of advances in percutaneous and transcatheter embolotherapy and sclerotherapy. Preprocedure evaluation of a malformation with either ultrasound or MRI is helpful not only in defining the extent of the abnormality but also in aiding to determine possible direct access into the lesion. Venous malformations may have a number of large-diameter connections to the deep venous system. It is important to control the injection to prevent the liquid embolics from entering these central veins.

Keywords

vascular malformations
hemangiomas
arteriovenous malformations
transcatheter embolotherapy
sclerotherapy

TABLE 25.1 Abbreviated International Society for the Study of Vascular Anomalies Classification for Vascular Anomalies

Vascular Anomalies				
Vascular Tumors			Vascular Malformations	
Benign	**Locally Aggressive**	**Malignant**	**Simple**	**Combined**
Infantile hemangioma	Kaposiform hemangioendothelioma	Angiosarcoma	Capillary malformation (CM)	CVM, CLM
Congenital hemangioma	Retiform hemangioendothelioma	Epithelioid hemangioendothelioma	Lymphatic malformation (LM)	LVM, CLVM
Tufted hemangioma	PILA, Dabska tumor		Venous malformation (VM)	CAVM
Spindle-cell hemangioma	Composite hemanigoendothioma		Arteriovenous malformation (AVM)	CLAVM
Epithelioid hemangioma	Kaposi sarcoma		Arteriovenous fistula	
Pyogenic granuloma				

CAVM, Capillary-arteriovenous malformation; *CLAVM,* cerebral-lymphatic-arteriovenous malformation; *CLM,* capillary-lymphatic-malformation; *CLVM,* capillary-lymphatic-venous malformation; *CVM,* capillary-venous-malformation; *LVM,* lymphatic-venous-malformation.

Abbreviated ISSVA classification for Vascular Anomalies by International Society for the Study of Vascular Anomalies is licensed under a Creative Commons Attribution 4.0 International License.

within the first 2 to 3 weeks of life. They are the most common, benign tumor of infancy. There is a subgroup of hemangiomas that present fully formed at birth known as congenital hemangiomas. These congenital hemangiomas (as opposed to infantile hemangiomas) do not exhibit the expected accelerated postnatal growth. Some congenital hemangiomas involute quickly over the first year of life and are called *rapidly involuting congenital hemangiomas* (RICHs), whereas some may persist indefinitely without treatment and are called *noninvoluting congenital hemangiomas* (NICHs).[2]

The majority of infantile hemangiomas are localized and, although disconcerting to parents and care providers, are nonthreatenting.[3] For these lesions, observation and routine monitoring by the pediatrician or dermatologist are acceptable treatment options. A minority of infantile hemangiomas can, however, cause significant morbidity. These require early recognition, timely referral to a specialist, and prompt intervention to minimize complications. Worrisome presentations include multiple hemangiomas and sensitive locations such as beard distribution, periocular, perioral, nasal tip, large regions of the face and neck, and the lumbosacral spine region. In general, larger hemangiomas located on the face are more likely to require treatment. One of the strongest indications for the use of the laser is the presence of ulceration. Other symptoms necessitating therapeutic intervention include congestive heart failure, airway obstruction, dysphagia, infection, failure to thrive, external auditory canal occlusion, visual axis impairment, and severe facial deformity.

Hemangiomas in adults are not hemangiomas; a hemangioma is a childhood-only birthmark. If a vascular birthmark was not present in the childhood stage, then it is not a hemangioma. Hemangiomas are not arteriovenous malformations (AVMs), and vice versa. Although there is arterial inflow that identifies both these lesions, hemangiomas demonstrate typical tumor vascularity with a central arterial pedicle, and there is fairly minimal, if any, shunting identified in the outflow vessels. An AVM, on the other hand, may have significant venous shunting, with resultant low-resistance arterial inflow and arterialized pulsatility in the venous outflow vessels. Venous enlargement is also common in AVMs, resulting from the pressurized, shunted, arterial flow into the nidus.

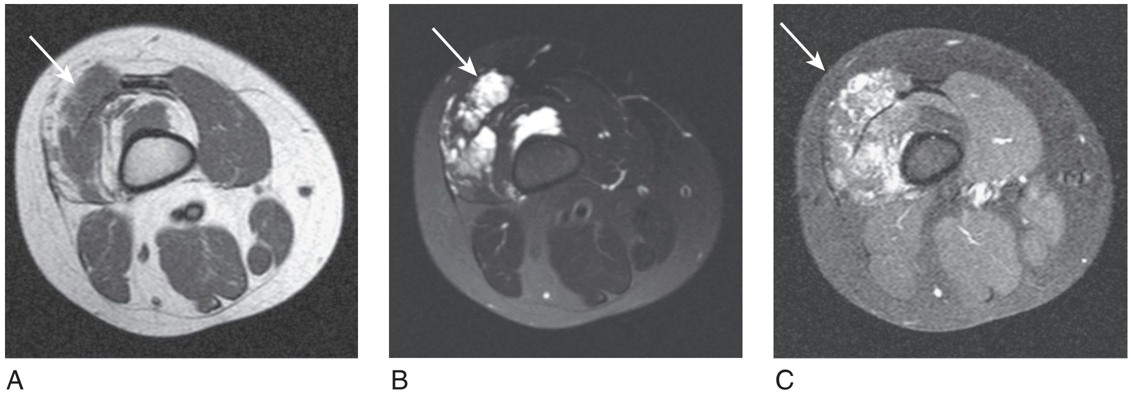
Muscular skeletal hemangiomas are not hemangiomas. They are venous malformations that occur in the muscle. With the correct clinical history, a properly performed magnetic resonance imaging (MRI) examination should be almost pathognomonic. Vertebral body hemangiomas are also not hemangiomas; they are venous malformations that occur in bone. Review of the literature demonstrates numerous studies showing "increased vascularity" of these lesions. However, the increased vascularity is from the venous pooling in the lesion and not from arterial hypertrophy and neovascularity. Liver hemangiomas are not hemangiomas; they are venous malformations occurring in the liver.

VASCULAR MALFORMATIONS

Vascular malformations are malformed or dysplastic embryologic spaces that are characterized by normal endothelial cell turnover and abnormal vascular anatomy characterized by the dysplastic vessels involved. All vascular malformations can be placed into one of three groups: low flow, high flow, and combined.

Low-Flow Vascular Malformations—Venous Malformations

Low-flow venous malformations are the most common form of vascular malformation; simplified, they are a number of tortuous vascular channels. They will be evident at birth and will grow with the child. These are common birthmarks present at birth, although they can be clinically occult if deep in location and usually do not become symptomatic until late childhood/early adolescence. Deep subcutaneous or intramuscular venous malformations often manifest with only local swelling and pain. The diagnosis can be difficult because extremity varicosities may be the only visible sign, especially in deep venous malformations. Superficial venous malformations can be seen with bluish-purple skin discoloration. Venous malformations can be divided into truncal and extratruncal lesions. On physical examination, these lesions are soft and easily compressible and will often demonstrate engorgement, especially when the affected extremity is placed in a dependent position. Extratruncal lesions occur from remnants of primitive vessels early in development and are usually dysplastic and diffusely infiltrative. They commonly involve the deep soft tissue structures, but patients may have constant pain. Truncal venous malformations occur in differentiated and later-stage vascular structures and patients may have an impressive cutaneous manifestation with limb swelling and varicosities. Diffuse malformations involve multiple areas or regions and are usually part of a syndrome such as Klippel-Trenaunay syndrome. Diagnosis of venous malformations can usually be made by clinical examination. However, imaging studies, such as MRI, are performed to evaluate the extent and guide treatment. On MRI, venous malformations



■ **Fig. 25.1** A 10-year-old boy with focal thigh swelling and pain. Axial T1-weighted (A), T2-weighted (B), and T1-weighted after gadolinium (C) images demonstrating a T1 isointense and T2 hyperintense signal and enhancement after gadolinium in a venous malformation *(arrow)* in the lateral aspect of left lower extremity.

are T1 isointense to muscle and T2 hyperintense and demonstrate late enhancement (Fig. 25.1).

Low-Flow Vascular Malformations—Lymphatic Malformations

These lesions are abnormalities of lymphatic etiology that often appear in infancy or early childhood, with more than 90% of patients presenting by age 2 years. More than 75% of these lesions are seen in the craniocervical region. Patients with these lesions usually present for treatment earlier than those with venous malformations, secondary to the cosmetic concerns of the localized mass effect or swelling caused by the lesions. Lymphatic malformations (LMs) will often fluctuate in size secondary to trauma, inflammation, or intralesional hemorrhage. Antibiotics are administered for fever or erythema. Unlike venous malformations, LMs are not usually painful. Larger LMs, however, can result in airway obstruction, speech abnormalities, and dysphagia. In the past, these lesions were called lymphangiomas or cystic hygromas. These outdated terms should no longer be used in clinical practice.

There are two subtypes of LMs: macrocystic LM and microcystic LM. Because the imaging characteristics and treatment options are different, it is important to recognize these subtypes. Macrocystic LMs appear similar to venous malformations on MRI except that they only demonstrate minimal, if any, peripheral enhancement. Macrocystic LMs are often easily accessible for sclerotherapy treatments. On the other hand, microcystic LMs are often only observed, with treatment options usually limited to surgical debulking and percutaneous management of cutaneous complications such as recurrent cellulitis or sclerotherapy treatment of bleeding superficial vesicles.

Low-Flow Vascular Malformations—Capillary Malformations

Capillary malformations (CMs) are capillary dilatations. They are characterized by ectatic papillary dermal capillaries and postcapillary venules in the upper reticular dermis and were commonly referred to as “port-wine” stains. CMs are present at birth and grow in size commensurate with the child because they have no tendency toward involution. At birth, CMs are usually pink or reddish and will usually darken with advancing age. They present as solitary lesions but can be associated with other vascular malformations. A typical worrisome location is the midline lower back region because CMs in this location may be associated with tethered spinal cord. Facial CMs may be associated with Sturge-Weber syndrome. Other conditions associated with CMs include Klippel-Trenaunay syndrome and Proteus syndrome.

High-Flow Vascular Malformations—Arteriovenous Fistulas

Arteriovenous fistulas (AVFs) are an abnormal connection between an artery and a vein. Although these high-flow lesions can be congenital, often they are acquired as the result of surgery, penetrating trauma, or erosion by adjacent disease processes. As blood follows the path of least resistance, flow in both the inflow artery and outflow vein increases. The resistance in the fistula can be low enough that the fistula tract causes a "steal phenomenon" to arterial supply distally and can actually cause a reversal of arterial flow in the distal arterial segment. This "parasitic circulation" can result in decreased arterial pressures in the distal capillary beds and cause tissue ischemia and pain. Treatment is usually from an endovascular approach, but this depends on the location of the AVF.

High-Flow Vascular Malformations—Arteriovenous Malformations

AVMs are abnormal connections between an artery and a vein that have a tortuous segment of dysplastic vessels (called the nidus) between the supplying artery or arteries and the draining vein or veins. Usually, these lesions are asymptomatic, but they can have varied symptoms, including heart failure, neuropathy, pain, or bleeding. Symptomatic, small, and superficial AVMs are occasionally treated with surgical resection; however, most AVMs are inoperable because they are large and diffuse in nature and involve important normal adjacent structures. With improvement of catheter technology, super-selective techniques, and the use of liquid embolic agents, embolotherapy has emerged as the primary mode of treatment for the management of peripheral AVMs.

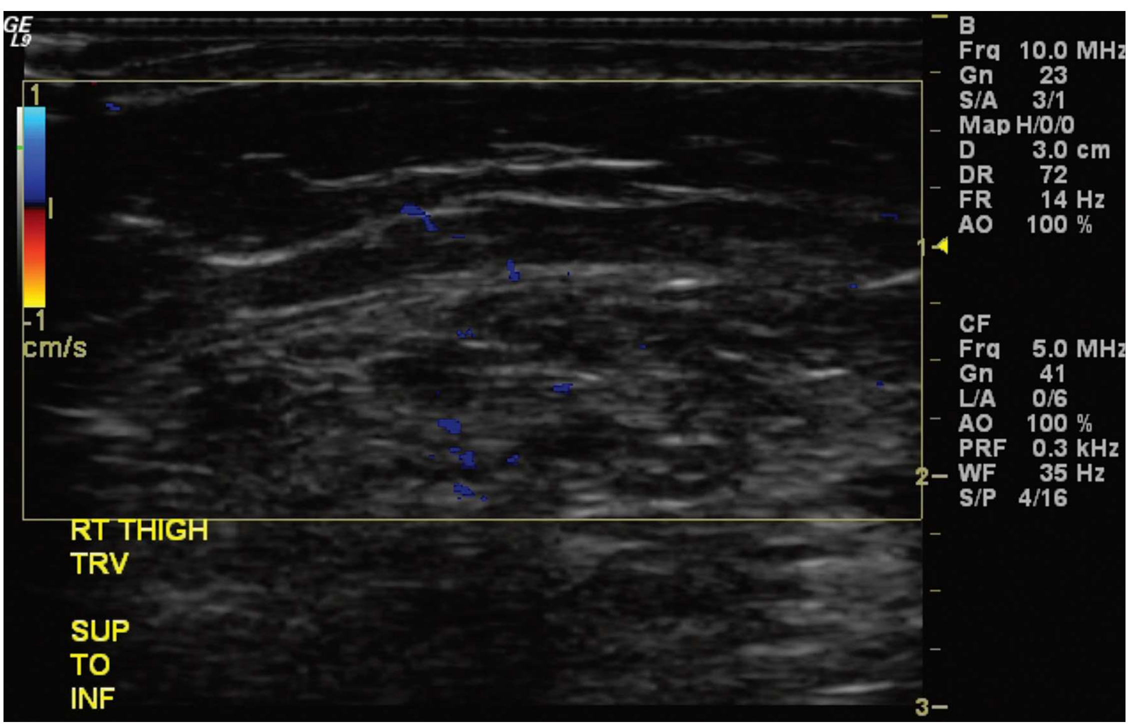
YAKES ARTERIOVENOUS MALFORMATION CLASSIFICATION

The Yakes classification is a relatively new system that characterizes AVMs based on their angioarchitectures suggesting possible treatment strategies. A type I AVM is a direct arteriovenous fistula that can be occluded with mechanical devices such as coils or plugs. A type IIa AVM has multiple inflow arteries into a traditional central nidus with direct artery to arteriolar and vein to venule communication. A type IIb AVM similarly has multiple inflow arteries, which supplies a characteristic nidus that shunts into an enlarged draining vein. Type II AVM therapy should focus on sclerosing the nidus, either through the arterial inflow vessel or direct injection. A type IIIa AVM features multiple artery-to-arteriole communications into an enlarged central vein where the vein wall serves as the nidus. Its treatment includes transarterial embolization of the nidus along with possibly direct puncture and occlusion of the venous outflow vessel. A type IIIb AVM features multiple artery to arterioles communications into an enlarged central vein with multiple, usually dilated, outflow veins. Its treatment includes a strategy of transarterial embolization of the inflow as well as puncture and occlusion of the enlarged dominant venous outflow along with the occlusion of the other outflow veins with coils or plugs. A type IV AVM has multiple small arteriolar to venule communications that diffusely infiltrate tissues that should be addressed with superselective ethanol (50%) injection via direct injection or transcatheter approach.

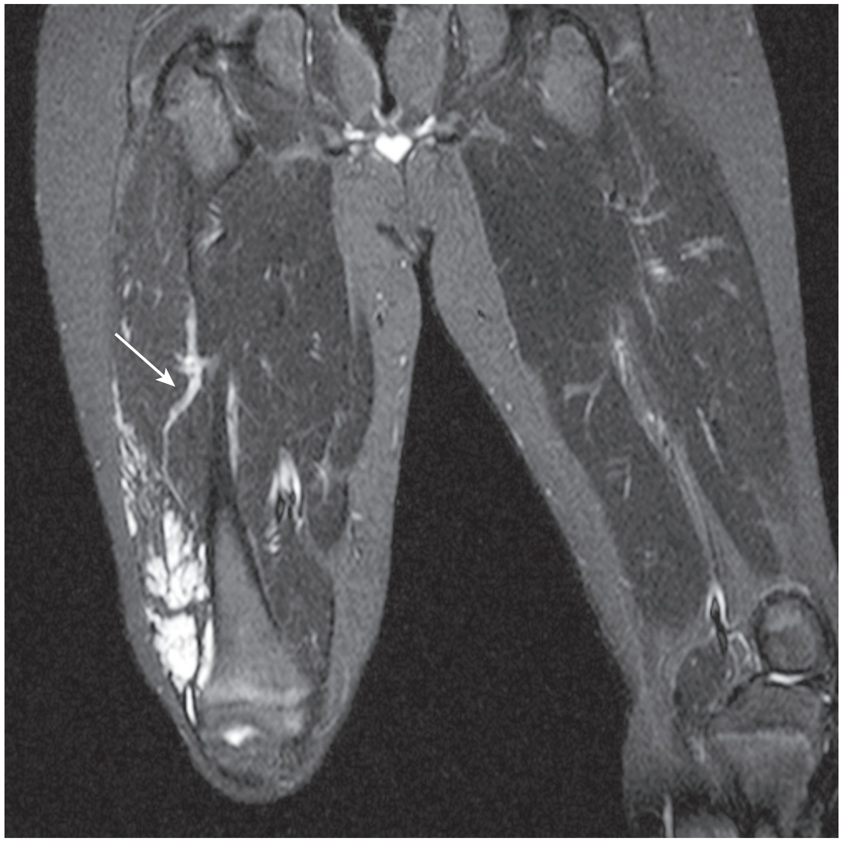
TREATMENT OF VENOUS MALFORMATIONS

Treatment of venous malformations has improved over the past decade as a result of advances in percutaneous and transcatheter embolotherapy and sclerotherapy. Localized and diffuse venous malformations have been treated with sclerotherapy, whereas some localized lesions may be resected. Sclerotherapy can be performed with ethanol, liquid, and foam detergents. In our practice, we treat venous malformations with ethyl alcohol or sodium tetradecyl sulfate (STS) sclerotherapy. When dehydrated ethyl alcohol is used, however, the potential for serious complications is increased by an order of magnitude. Local ethanol complications are related to the transmural necrosis of the agent and spread to the surrounding tissues.

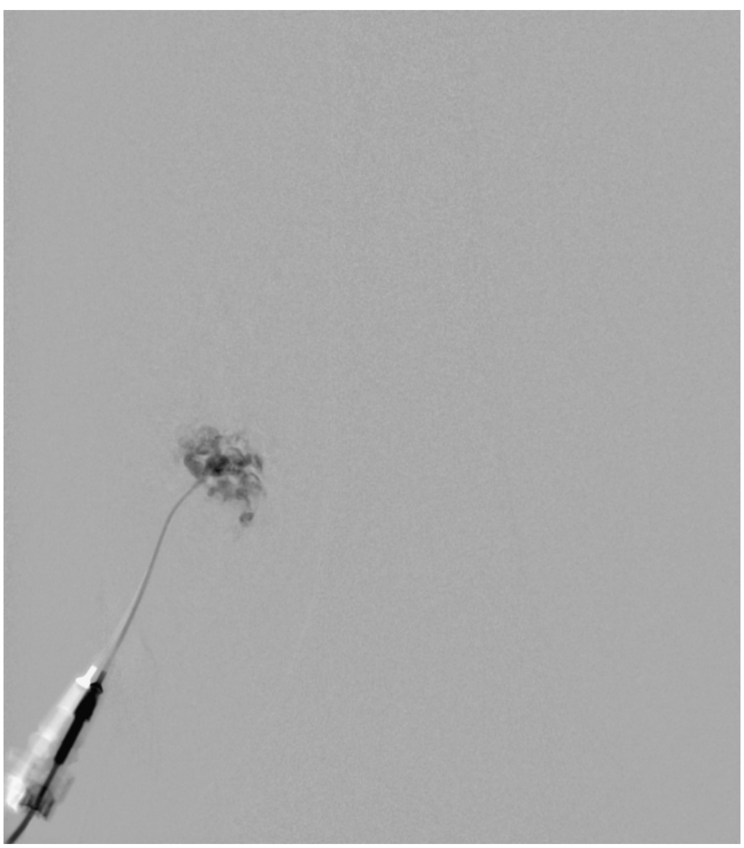
Preprocedure evaluation of a malformation with either ultrasound or MRI is helpful not only in defining the extent of the abnormality but also in aiding practitioners to determine possible direct access into the lesion (Figs. 25.2 and 25.3). Access into the malformation is usually performed with ultrasound and fluoroscopic guidance (Fig. 25.4). Biplane fluoroscopy can be used if the location of the lesion allows. Direct comparison with the magnetic resonance images can also be very helpful. Proper positioning on the angiographic table is critical to successfully access the malformation and to allow proper



■ **Fig. 25.2** Ultrasound Doppler examination demonstrates a heterogeneous lesion with no increased flow consistent with a low-flow vascular venous malformation.



■ **Fig. 25.3** Coronal fat-saturated T1-weighted image. After the addition of gadolinium, enhancement of the dysplastic draining vein is demonstrated *(arrow)*.



■ **Fig. 25.4** Initial venogram with dilute contrast.

anesthesia monitoring. All alcohol embolization procedures are performed with the patient under general anesthesia to allow for proper sedation and pain management, as well as to prepare should complications of treatment (such as cardiovascular collapse) occur. In addition, a tourniquet for control with a calibrated cuff can be used if the location of the lesions allows. When using a tourniquet, we do not exceed diastolic blood pressure and often only use minimal pressures in the 20 to 40 mm Hg range.

The technique for alcohol administration has evolved. Previously, dehydrated ethanol was opacified with a small amount of ethiodized oil (Ethiodol) to increase the visualization of the administered alcohol under fluoroscopic control. Currently, we use a negative contrast technique. This consists of opacification of the malformation with contrast, followed by administration of the ethyl alcohol that replaces the contrast on live fluoroscopic evaluation (Fig. 25.5). Once we have either treated the entire lesion or reached our sclerosant limits, the procedure is concluded (Fig. 25.6). The absolute maximum for ethyl alcohol is 1 mL per kg. In our practice, we rarely exceed 0.5 mL per kg in one procedure. A good working dose limit for ethyl alcohol is 0.25 mg per kg for the entire case. We also limit our total injected volume to 0.1 mL per kg per injection and allow at least 5 minutes between alcohol administrations. By adhering to these recommendations, we have completed safe treatment for many patients, without the use of pulmonary arterial catheter monitoring. Similarly, STS can be injected as a liquid or as a foam. (Fig. 25.7) The foam is more viscous and slightly easier to control. It can be applied via a transcatheter or direct puncture approach. A negative contrast technique is used during the injection (Figs. 25.8–25.10).

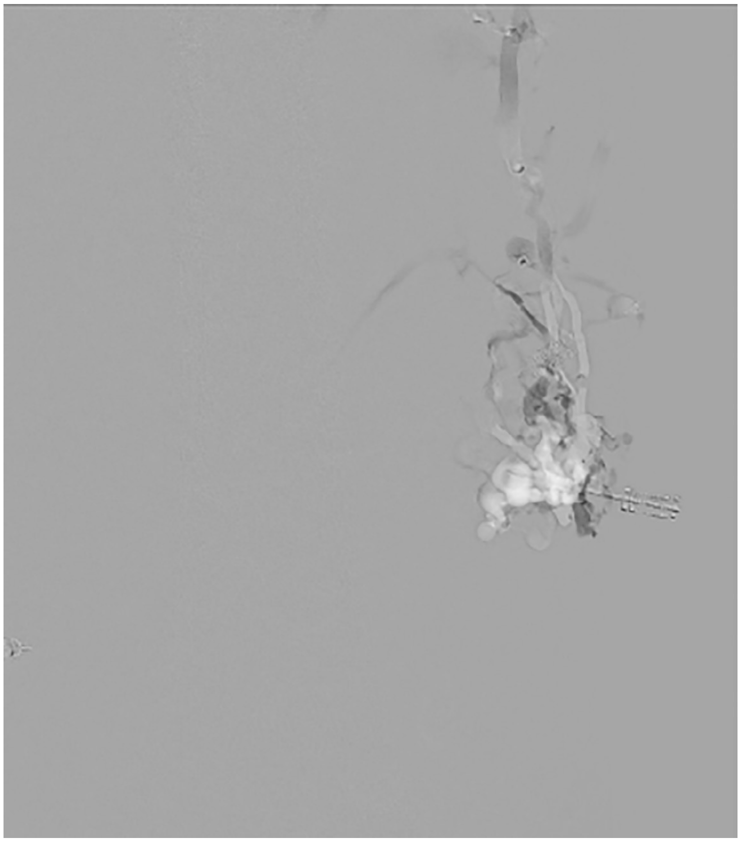
■ **Fig. 25.5** Direct puncture venography using a small angiocatheter.

■ **Fig. 25.6** Posttreatment scout film demonstrating two separate treatment punctures.

■ **Fig. 25.7** Syringe of 3 mL of 3% sodium tetradecyl sulfate used to create foam.

■ **Fig. 25.8** A 22-year-old male with a left thigh painful venous malformation (Yakes IIIa). The patient underwent direct injection of the nidus vessels. The dominant outflow vein was later closed.

■ **Fig. 25.9** Direct venogram demonstrating filling of the nidus vessels.



■ **Fig. 25.10** Images after the injection of sodium tetradecyl sulfate foam using negative-contrast technique.

Occasionally, a lesion can be treated in one session; however, most require several (two to four) treatments to reach our clinical endpoint of treating the pain associated with a symptomatic malformation. We do not treat asymptomatic lesions unless there is a significant cosmetic issue, which can occur with lesions on the face and neck. Treatments are usually spaced apart by 3 to 4 months to allow for postprocedural edema to resolve.

Our standard postprocedure orders include intravenous ketorolac, dexamethasone, and antibiotic coverage for the first 24 hours of treatment. Following overnight observation, patients are discharged home with an outpatient oral medication regimen of a methylprednisolone dose pack, ibuprofen therapy, and oral antibiotics.

Evaluation of the venous drainage of venous malformations is essential in their treatment. Malformations with no or limited venous drainage can be treated with embolic agents with higher effectiveness and less risk than malformations with normal draining vessels, whereas lesions with drainage into dysplastic veins or a venous ectasia can be most problematic to treat. Venous malformations may have a number of large-diameter connections to the deep venous system. It is important to control the injection to prevent the liquid embolics from entering these central veins. Burrows et al. have reported good result in 75% to 90% of their patient cohort treated with serial alcohol administration. However, many patients have chronic pain symptoms, and their clinical improvement may lag significantly behind a more technically successful treatment.

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